

# Project Report

## Import Data using Transform Maps (Spreadsheet)

Team Lead: Mohammedmoinuddin S    Date: 29 September 2026

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## 1. INTRODUCTION

### Team Members

| Team Member         | Role      | Responsibility                                           |
|---------------------|-----------|----------------------------------------------------------|
| Mohammedmoinuddin S | Team Lead | Planning, Coalesce setup, review and sign-off            |
| Daniel Giftson R    | Member    | Spreadsheet preparation and Load Data (import set table) |
| Mizwar Shahul R     | Member    | Employee Test table, transform map and field mapping     |
| Adnan Mustafa A     | Member    | Transform run, validation, re-import and UAT testing     |
| Akshay Kumar G      | Member    | Reports and Employee Analytics dashboard                 |

### 1.1 Project Overview

This micro project imports structured employee data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. It simulates a real-world scenario in which bulk employee data arrives in Excel format and must be migrated efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which acts as a staging area; a Transform Map then maps source fields to the target table fields. When the transformation runs, records are created in the target table and validated for integrity. The project covers data loading, field mapping, transformation, coalesce-based duplicate prevention, and reporting.

### 1.2 Purpose

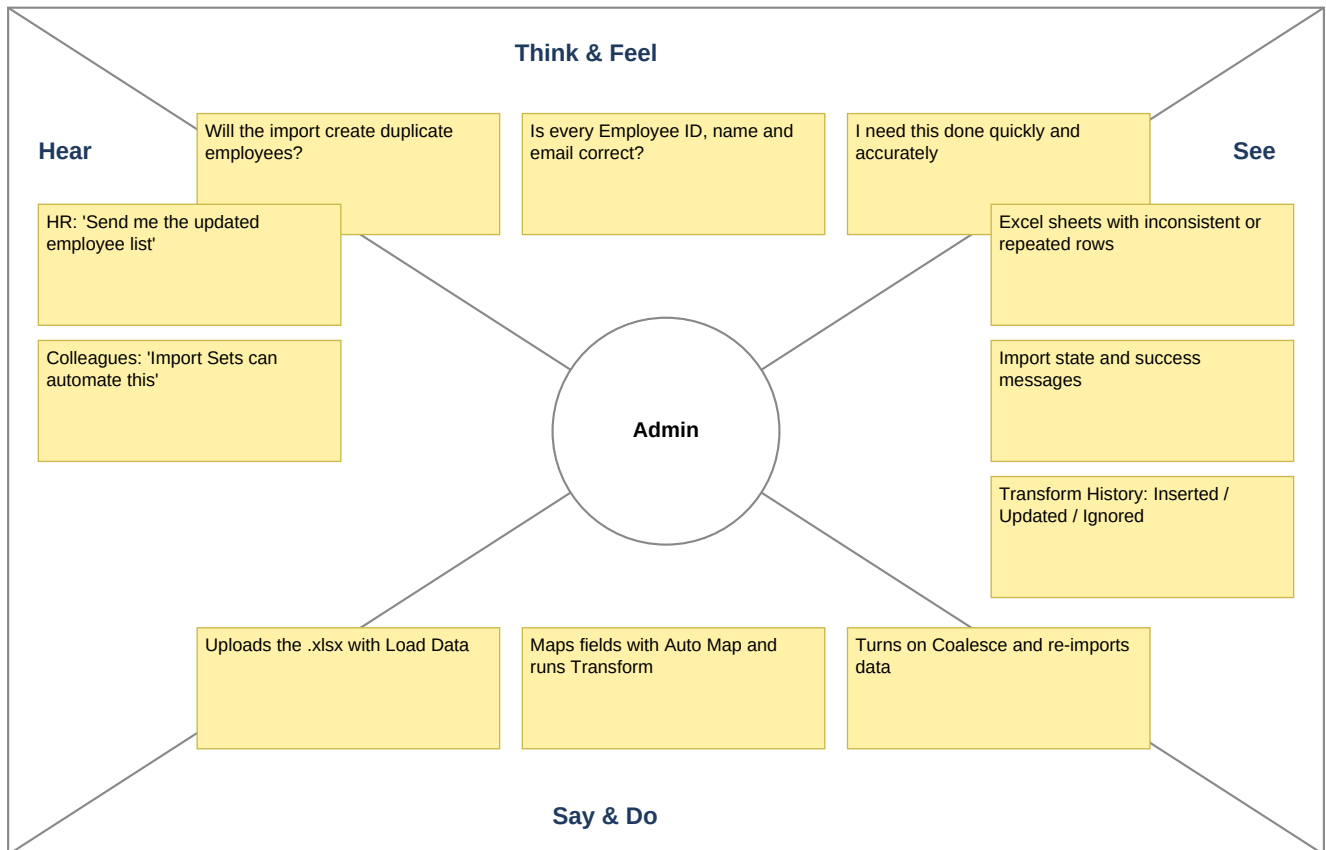
To help beginners understand the end-to-end data import process in ServiceNow, and to deliver an automated employee data management solution that keeps data accurate, duplicate-free and visible through reports and a dashboard. Data handled: Employee ID, Employee Name, Email, Department, Location.

## 2. IDEATION PHASE

### 2.1 Problem Statement

| Problem Statement (PS) | I am (Customer)                                              | I'm trying to                                                                                                 | But                                                                                                 | Because                                                                                                  | Which makes me feel                                            |
|------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| PS-1                   | A ServiceNow administrator handling employee onboarding data | Bulk-load employee records (ID, name, email, department, location) received in an Excel sheet into ServiceNow | Entering records one by one is slow, and uploading the same sheet again creates duplicate employees | There is no automated field mapping or unique-key check between the spreadsheet and the ServiceNow table | Frustrated and worried about the accuracy of the employee data |
| PS-2                   | An HR / operations team member                               | See how many employees belong to each department and location                                                 | The data sits in spreadsheets and has to be counted manually                                        | There are no reports or a dashboard built on the imported employee table                                 | Uninformed and slow to make staffing decisions                 |

## 2.2 Empathy Map Canvas



### Pain

- Manual entry of many employee rows is slow and error-prone
- Duplicate employees when the same data is imported twice
- Wrong or unmapped fields between sheet and table
- No visual summary of employee data

### Gain

- Fast bulk import through Import Sets and Transform Maps
- Clean data - Coalesce updates instead of duplicating
- Clear audit of Inserted / Updated / Ignored rows
- Department and location insights on one dashboard

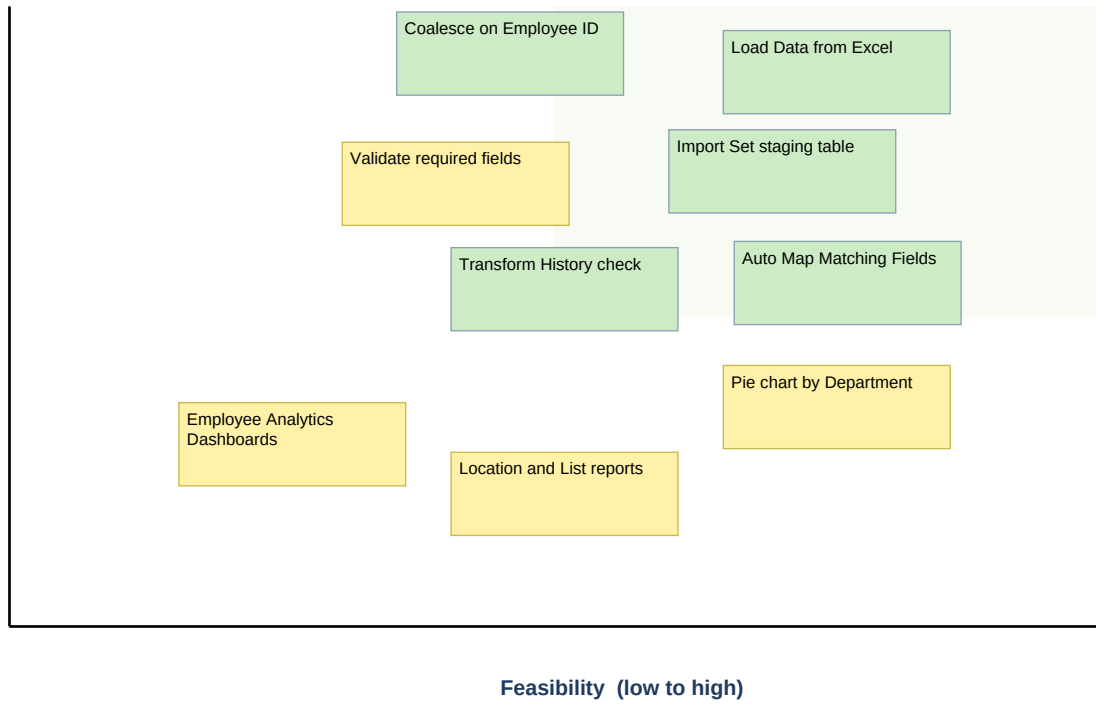
## 2.3 Brainstorming

Step-1: team gathering and selection of PS-1. Step-2: ideas grouped into Data Import, Data Integrity and Insights. Step-3: ideas ranked by importance and feasibility.

| Mohammedmoinuddin S         | Daniel Giftson R             | Mizwar Shahul R            | Adnan Mustafa A             | Akshay Kumar G                     |
|-----------------------------|------------------------------|----------------------------|-----------------------------|------------------------------------|
| Plan milestones and sprints | Prepare .xlsx sample sheet   | Create Employee Test table | Validate ID, Name, Email    | Pie chart by Department            |
| Coalesce on Employee ID     | Load Data into staging table | Transform map and Auto Map | Ignore identical re-imports | Location and Employee List reports |
| Review Transform History    | Create Employee Import table | Run Transform and verify   | Personalize list columns    | Employee Analytics Dashboards      |

Importance (low to high)

*Do first: high importance, high feasibility*



### 3. REQUIREMENT ANALYSIS

#### 3.1 Customer Journey Map

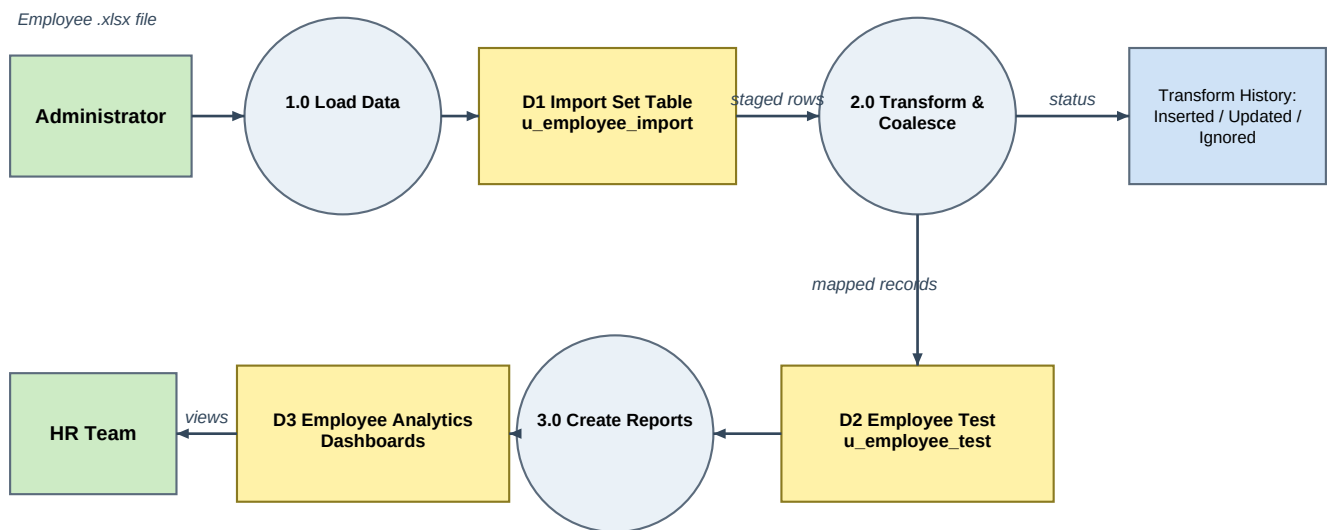
The administrator's journey follows five stages: Entice (receive the employee sheet), Enter (prepare the sheet and table), Engage (load, map and transform), Exit (verify records) and Extend (coalesce re-imports, reports and dashboard). The full map is provided in the separate Customer Journey Map document.

#### 3.2 Solution Requirement

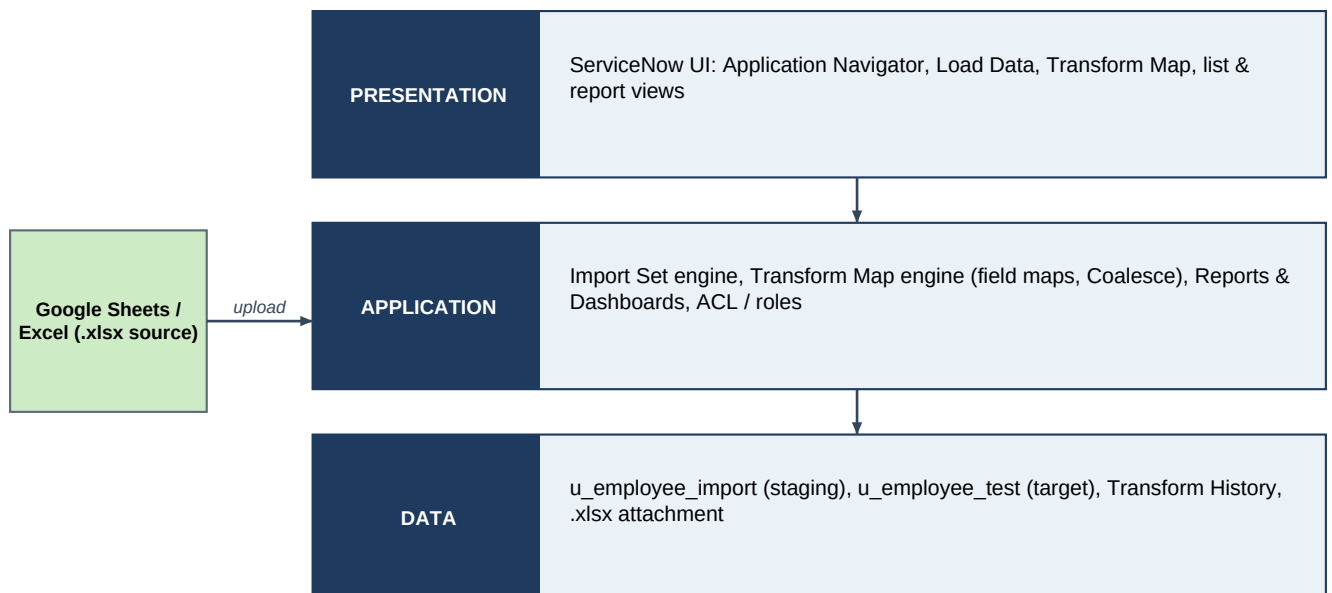
| FR No. | Functional Requirement (Epic)    | Sub Requirement (Story / Sub-Task)                                                                                                                                |
|--------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | Spreadsheet Preparation          | Create a sample sheet with Employee ID, Employee Name, Email, Department, Location<br>Download it in .xlsx format (Sample Spreadsheet.xlsx)                       |
| FR-2   | Target Table Creation            | Create custom table Employee Test (u_employee_test)<br>Add five String fields through Form Layout and verify them on the form                                     |
| FR-3   | Import Set (Staging) Table       | Use Load Data to create Employee Import (u_employee_import)<br>Upload the .xlsx file (sheet 1, header row 1) and generate the transform map                       |
| FR-4   | Field Mapping (Transform Map)    | Create transform map Sample Spreadsheet Import (source: Employee Import, target: Employee Test)<br>Use Auto Map Matching Fields and confirm source-target mapping |
| FR-5   | Data Transformation & Validation | Run Transform and confirm the success message<br>Validate that Employee ID, Name and Email hold valid data; verify records in Employee Test                       |
| FR-6   | Duplicate Prevention (Coalesce)  | Set Coalesce = true on Employee ID in the field map<br>Existing IDs are updated; new IDs are inserted; identical rows are ignored                                 |
| FR-7   | Reports & Dashboard              | Create reports: Employees by Department (pie chart), Employees by Location, Employee List<br>Add them to the Employee Analytics Dashboards dashboard              |

| NFR No. | Non-Functional Requirement | Description                                                                                                                                                                                                    |
|---------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1   | Usability                  | Administrators complete the import through standard ServiceNow modules (Load Data, Transform Maps, Reports) with no coding; success messages confirm each step.                                                |
| NFR-2   | Security                   | Only users with administrative/development access can import data. Dashboard creation is governed by the pa_dashboards ACL (Admin Overrides), and report sharing can be restricted by users, groups and roles. |
| NFR-3   | Reliability                | The staging table isolates raw data from the target table; Transform History records Inserted, Updated and Ignored rows so every run can be audited.                                                           |
| NFR-4   | Performance                | Each import and transform completes with state Completed and Success; repeated imports of unchanged data are ignored instead of re-processed into new records.                                                 |
| NFR-5   | Availability               | Runs on a ServiceNow instance (Personal Developer Instance or an instance with development access); imports can be re-run at any time.                                                                         |
| NFR-6   | Scalability                | The same import table and transform map are reused for any number of uploads; more fields and rows can be added by extending the table and field maps.                                                         |

### 3.3 Data Flow Diagram



### 3.4 Technology Stack



Platform: ServiceNow (PDI). Components: Load Data and Import Sets, Transform Map with Coalesce, staging table u\_employee\_import, target table u\_employee\_test, Reports and Dashboards; source data prepared in Google Sheets / Excel (.xlsx). No custom code or open-source frameworks are used.

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

**Customer segment:** ServiceNow administrators and HR teams. **Problem:** slow, duplicate-prone manual entry of spreadsheet data. **Root cause:** no automated field mapping or unique-key check. **Solution:** Import Set staging table + Transform Map + Coalesce on Employee ID + reports. **Emotion:** from frustrated and worried to confident and in control. The full canvas is in the Problem-Solution Fit document.

### 4.2 Proposed Solution

| S.No. | Parameter                             | Description                                                                              |
|-------|---------------------------------------|------------------------------------------------------------------------------------------|
| 1     | Problem Statement                     | Load Excel employee data into ServiceNow accurately, without manual entry or duplicates. |
| 2     | Idea / Solution                       | Staging table + transform map + Coalesce on Employee ID + reports and dashboard.         |
| 3     | Novelty / Uniqueness                  | No-code flow; Transform History shows inserted / updated / ignored rows per run.         |
| 4     | Social Impact / Customer Satisfaction | Saves time, reduces errors, keeps employee data trustworthy.                             |
| 5     | Business Model                        | Internal efficiency solution on an existing ServiceNow licence.                          |
| 6     | Scalability                           | Reusable table and transform map; extendable fields, targets and schedules.              |

### 4.3 Solution Architecture

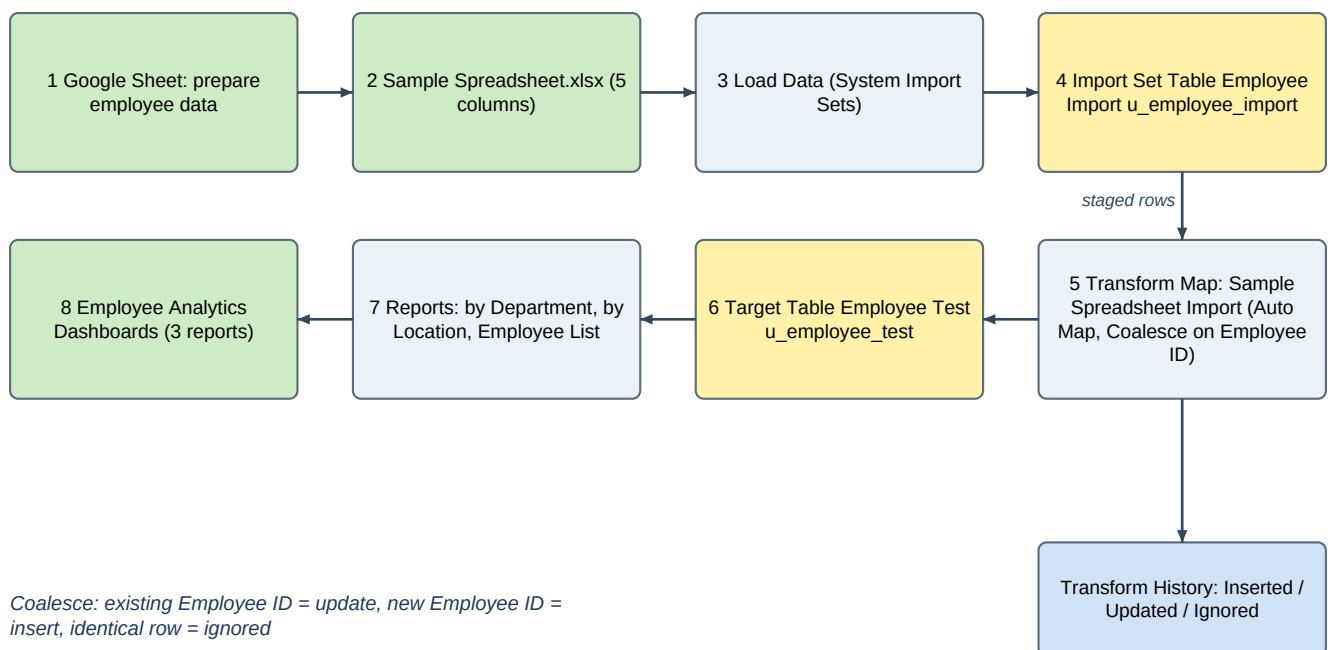


Figure: Architecture and data flow of the spreadsheet import solution

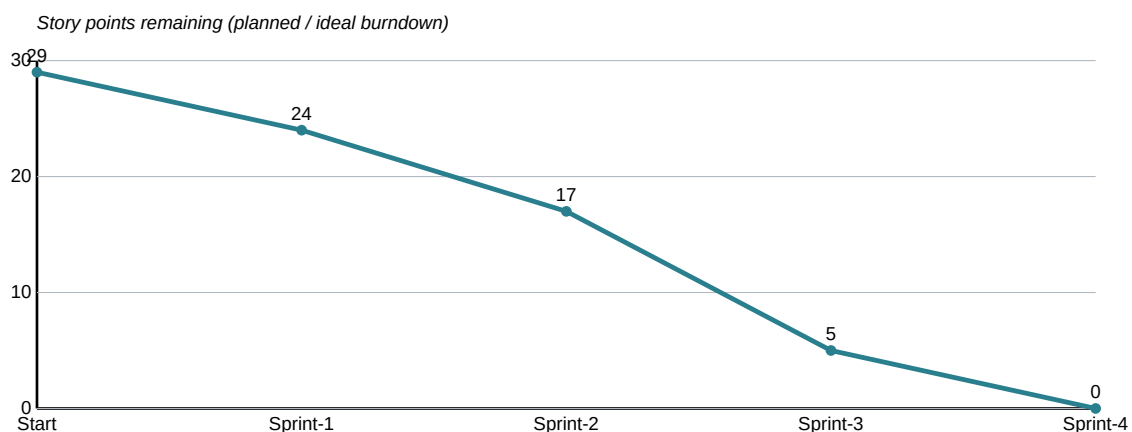
## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                | Story Points | Priority | Team Members        |
|----------|-------------------------------|-------------------|------------------------------------------------------------------|--------------|----------|---------------------|
| Sprint-1 | Spreadsheet Preparation       | USN-1             | Create the employee spreadsheet and download it as .xlsx         | 2            | High     | Daniel Giftson R    |
| Sprint-1 | Target Table Creation         | USN-2             | Create Employee Test table with five String fields               | 3            | High     | Mizwar Shahul R     |
| Sprint-2 | Import Set Table              | USN-3             | Create Employee Import table using Load Data                     | 2            | High     | Daniel Giftson R    |
| Sprint-2 | Field Mapping                 | USN-4             | Create transform map and Auto Map Matching Fields                | 3            | High     | Mizwar Shahul R     |
| Sprint-2 | Data Transformation           | USN-5             | Run Transform and verify records in Employee Test                | 2            | High     | Adnan Mustafa A     |
| Sprint-3 | Data Validation               | USN-6             | Validate Employee ID, Name and Email                             | 2            | Medium   | Adnan Mustafa A     |
| Sprint-3 | Duplicate Prevention          | USN-7             | Enable Coalesce on Employee ID (est. 3h 20m)                     | 5            | High     | Mohammedmoinuddin S |
| Sprint-3 | Re-import of Data             | USN-8             | Import new Excel data and review Transform History (est. 3h 20m) | 5            | High     | Adnan Mustafa A     |
| Sprint-4 | Reports                       | USN-9             | Create three reports on Employee Test                            | 3            | Medium   | Akshay Kumar G      |
| Sprint-4 | Dashboard                     | USN-10            | Create dashboard and add the three reports                       | 2            | Medium   | Akshay Kumar G      |

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) |
|----------|--------------------|----------|-------------------|---------------------------|
| Sprint-1 | 5                  | 2 Days   | 29 Sep 2026       | 30 Sep 2026               |
| Sprint-2 | 7                  | 2 Days   | 01 Oct 2026       | 02 Oct 2026               |
| Sprint-3 | 12                 | 2 Days   | 05 Oct 2026       | 06 Oct 2026               |
| Sprint-4 | 5                  | 2 Days   | 07 Oct 2026       | 08 Oct 2026               |

Total story points = 29 over 4 sprints; velocity =  $29 / 4 = 7.25$  story points per sprint. Sprint dates and story points are planning estimates.



## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

| Test Case ID | Scenario                         | Test Steps                                                                                                                                                                    | Expected Result                                                             | Actual Result                                                                         | Pass/Fail |
|--------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------|
| FT-01        | Spreadsheet preparation          | Create a Google Sheet with the five columns and download it as .xlsx                                                                                                          | File downloads in .xlsx with Employee ID, Name, Email, Department, Location | Sheet prepared with employees SB0001 to SB0006                                        | Pass      |
| FT-02        | Target table and fields          | Tables > Create New (Employee Test, u_employee_test); add five String fields via Form Layout; Save                                                                            | All five fields are displayed on the Employee Test form                     | Fields displayed on the form                                                          | Pass      |
| FT-03        | Import set table creation        | Load Data > label Employee Import (u_employee_import); Submit; choose Create Transform Map                                                                                    | Success message; import table and transform map created                     | Success message displayed; transform map created                                      | Pass      |
| FT-04        | Field mapping                    | Open transform map Sample Spreadsheet Import; use Auto Map Matching Fields / Mapping Assist; Save                                                                             | Source and target fields are mapped                                         | Name to Employee Name, Email, Employee ID, Department, Location mapped (5 field maps) | Pass      |
| FT-05        | Transform execution              | Click Transform for import set ISET0010001; open Employee Test list                                                                                                           | State Complete, Success; records created in Employee Test                   | Complete, Success; 5 records created (SB0001 to SB0005)                               | Pass      |
| FT-06        | Coalesce on re-import            | Enable Coalesce in the field map; load the 6-row sheet again with Load Data; run Transform on ISET0010003                                                                     | Existing employees matched, new employee added, no duplicates               | Complete, Success; Employee Test holds 6 unique records (SB0001 to SB0006)            | Pass      |
| FT-07        | Verify records and list columns  | Open Employee Tests list; Personalize List Columns to show Employee ID, Name, Department, Email, Location                                                                     | All 6 records displayed with the selected columns                           | 6 of 6 records displayed with the selected columns                                    | Pass      |
| FT-08        | Report creation                  | Reports > New: Employees by Department (pie chart, group by Department, count); create Employees by Location and Employee List; Run and Save                                  | Three reports are created and display Employee Test data                    | Three reports created and saved                                                       | Pass      |
| FT-09        | Dashboard                        | Create Employee Analytics Dashboards (PA_DASHBOARDS.FORM after enabling Admin Overrides on the pa_dashboards Create ACL); add the three reports via Share > Add to Dashboards | Dashboard displays all three reports                                        | Three reports added to the dashboard                                                  | Pass      |
| PT-01        | Load Data run                    | Submit the Load Data form for the .xlsx file                                                                                                                                  | Import completes with State = Complete and Success                          | Complete, Success; processed 6, inserts 6, errors 0 (1.241 s)                         | Pass      |
| PT-02        | Transform run                    | Run Transform with the selected transform map for ISET0010003                                                                                                                 | Transform completes with State = Complete and Success                       | Complete, Success - Transformation complete                                           | Pass      |
| PT-03        | Repeated import - data integrity | Import the sheet again into the same staging table and transform map                                                                                                          | Employee Test record count reflects unique employees only                   | 6 unique Employee IDs; no duplicate employee records                                  | Pass      |

Result: 12 of 12 test cases passed. UAT defect count: 0 (see the UAT Report).



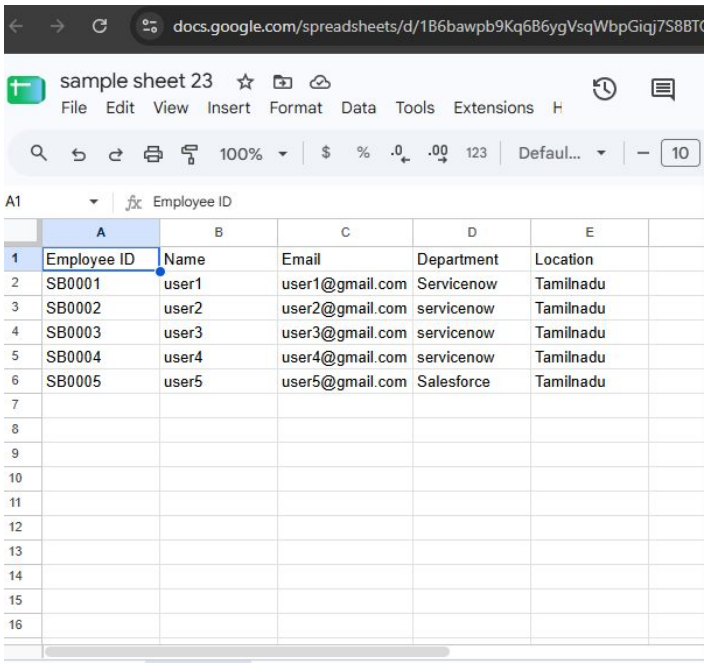
## 7. RESULTS

### 7.1 Output Screenshots

Screenshots from the ServiceNow instance are arranged in the order the project is carried out.

#### Milestone 1 - Creation of Spreadsheet and Table

Step 1: Sample spreadsheet created in Google Sheets with Employee ID, Name, Email, Department and Location (five employee rows).

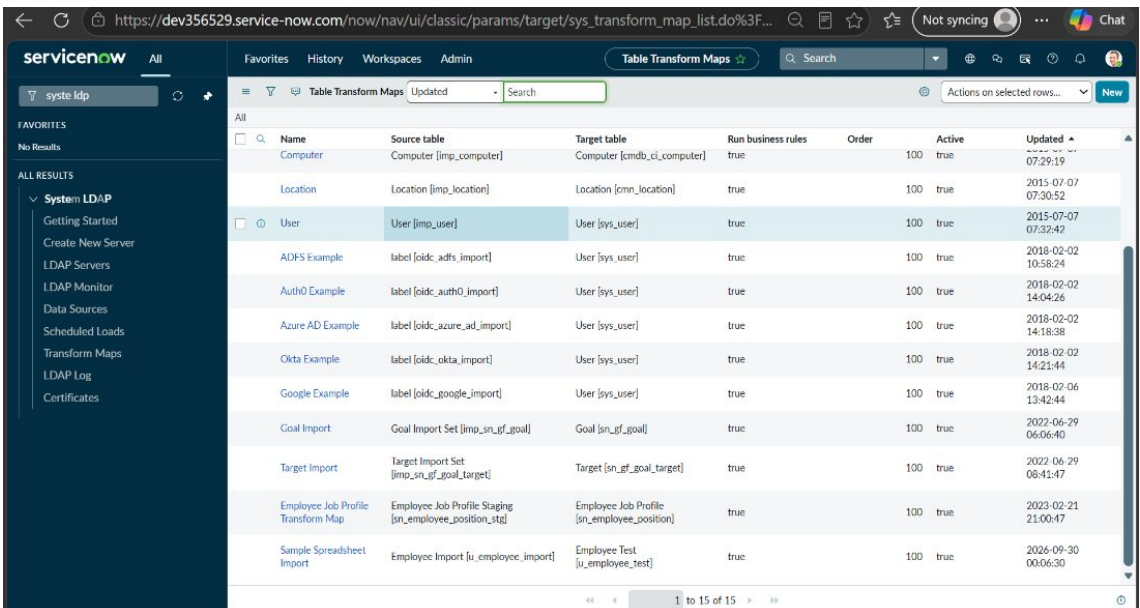


The screenshot shows a Google Sheet interface with a table containing employee data. The table has 5 columns: Employee ID, Name, Email, Department, and Location. The data is as follows:

| Employee ID | Name  | Email           | Department | Location  |
|-------------|-------|-----------------|------------|-----------|
| SB0001      | user1 | user1@gmail.com | ServiceNow | Tamilnadu |
| SB0002      | user2 | user2@gmail.com | ServiceNow | Tamilnadu |
| SB0003      | user3 | user3@gmail.com | ServiceNow | Tamilnadu |
| SB0004      | user4 | user4@gmail.com | ServiceNow | Tamilnadu |
| SB0005      | user5 | user5@gmail.com | Salesforce | Tamilnadu |

#### Milestone 2 - Import Set Table and Transform Map

Step 2: Table Transform Maps list showing Sample Spreadsheet Import (source Employee Import [u\_employee\_import], target Employee Test [u\_employee\_test]).



The screenshot shows the ServiceNow Table Transform Maps list. The table displays various import maps, including the 'Sample Spreadsheet Import' map. The data is as follows:

| Name                               | Source table                                            | Target table                                | Run business rules | Order | Active | Updated             |
|------------------------------------|---------------------------------------------------------|---------------------------------------------|--------------------|-------|--------|---------------------|
| Computer                           | Computer [imp_computer]                                 | Computer [cmdb_ci_computer]                 | true               | 100   | true   | 2015-07-07 07:29:19 |
| Location                           | Location [imp_location]                                 | Location [cmn_location]                     | true               | 100   | true   | 2015-07-07 07:30:52 |
| User                               | User [imp_user]                                         | User [sys_user]                             | true               | 100   | true   | 2015-07-07 07:32:42 |
| ADFS Example                       | label [oidc_adfs_import]                                | User [sys_user]                             | true               | 100   | true   | 2018-02-02 10:58:24 |
| Auth0 Example                      | label [oidc_auth0_import]                               | User [sys_user]                             | true               | 100   | true   | 2018-02-02 14:04:26 |
| Azure AD Example                   | label [oidc_azure_ad_import]                            | User [sys_user]                             | true               | 100   | true   | 2018-02-02 14:18:38 |
| Okta Example                       | label [oidc_okta_import]                                | User [sys_user]                             | true               | 100   | true   | 2018-02-02 14:21:44 |
| Google Example                     | label [oidc_google_import]                              | User [sys_user]                             | true               | 100   | true   | 2018-02-06 13:42:44 |
| Goal Import                        | Goal Import Set [imp_sn_gf_goal]                        | Goal [sn_gf_goal]                           | true               | 100   | true   | 2022-06-29 06:06:40 |
| Target Import                      | Target Import Set [imp_sn_gf_goal_target]               | Target [sn_gf_goal_target]                  | true               | 100   | true   | 2022-06-29 08:41:47 |
| Employee Job Profile Transform Map | Employee Job Profile Staging [sn_employee_position_stg] | Employee Job Profile [sn_employee_position] | true               | 100   | true   | 2023-02-21 21:00:47 |
| Sample Spreadsheet Import          | Employee Import [u_employee_import]                     | Employee Test [u_employee_test]             | true               | 100   | true   | 2026-09-30 00:06:30 |

Step 3: Transform map form - name, source table, target table and related links (Auto Map Matching Fields, Mapping Assist, Transform).

**Table Transform Map - Sample Spreadsheet Import**

Name: Sample Spreadsheet Import

Source table: Employee Import [u\_employee\_i...]

Target table: Employee Test [u\_employee\_test]

Field Maps (5):

| Source field  | Target field    | Coalesce |
|---------------|-----------------|----------|
| u_email       | u_email         | false    |
| u_name        | u_employee_name | true     |
| u_employee_id | u_employee_id   | false    |

Step 4: Mapping Assist - source fields mapped to Employee Test fields (Name to Employee Name, Email, Employee ID, Department, Location).

**Mapping Assist**

Source: Employee Import

Field Map:

| Source field | Target field  |
|--------------|---------------|
| Name         | Employee Name |
| Email        | Employee ID   |
| Employee ID  | Department    |
| Department   | Location      |
| Location     | Location      |

Data Viewer:

Employee Import

| Field       | Value               |
|-------------|---------------------|
| Comment     |                     |
| Created     | 2026-09-30 00:02:24 |
| Created by  | admin               |
| Department  | servicenow          |
| Email       | user3@gmail.com     |
| Employee ID | SB0003              |

Step 5: Field Maps (5) listing each source field, target field and Coalesce setting.

The screenshot shows the ServiceNow interface for configuring a Table Transform Map. The page title is "Table Transform Map - Sample Spreadsheet Import". The "Field Maps (5)" tab is selected, displaying a table with 5 rows of field mappings. The table has three columns: "Source field", "Target field", and "Coalesce".

| Source field  | Target field    | Coalesce |
|---------------|-----------------|----------|
| u_email       | u_email         | false    |
| u_name        | u_employee_name | true     |
| u_employee_id | u_employee_id   | false    |
| u_department  | u_department    | false    |
| u_location    | u_location      | false    |

Below the table, there are navigation controls showing "1 to 5 of 5" records.

Step 6: First transform run (ISET0010001) - State Complete, Completion code Success.

The screenshot shows the ServiceNow interface for a Transform Run. The page title is "ServiceNow". The "Progress" section shows the following details:

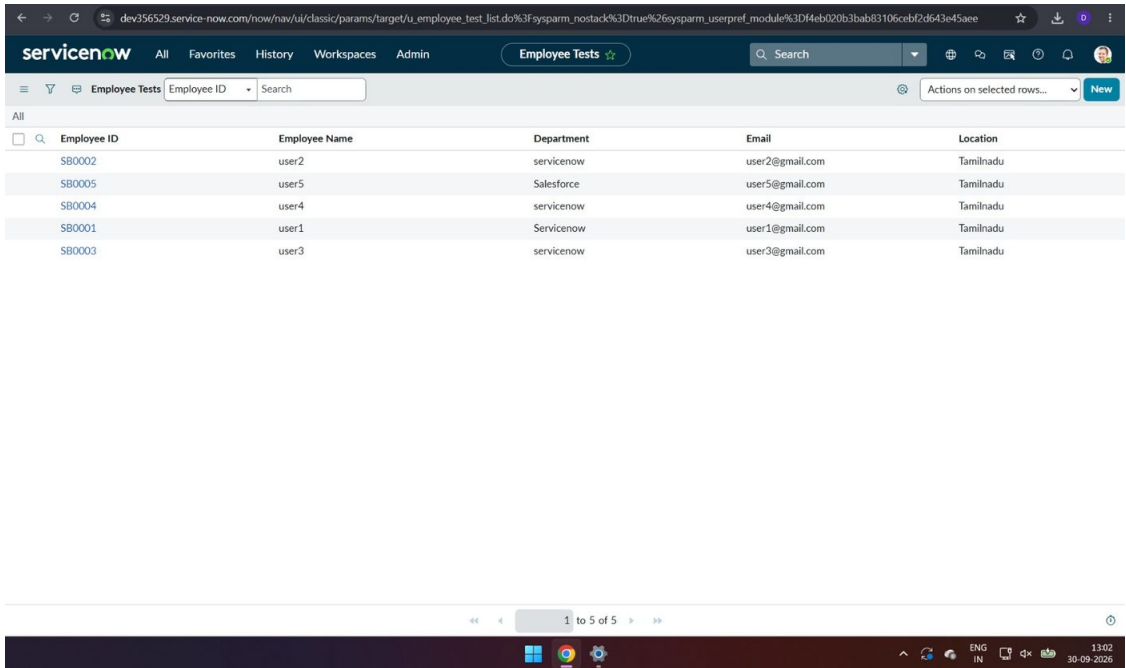
- Name: Transforming: ISET0010001
- State: **Complete**
- Completion code: **Success**
- Message: Transformation complete

The "Next steps..." section provides links for further actions:

- ISET0010001**: Go to the import sets for this data load
- Transform history**: Show the transform history, related errors and log
- Import log**: View the import log

### Milestone 3 - Transform, Validate and Coalesce

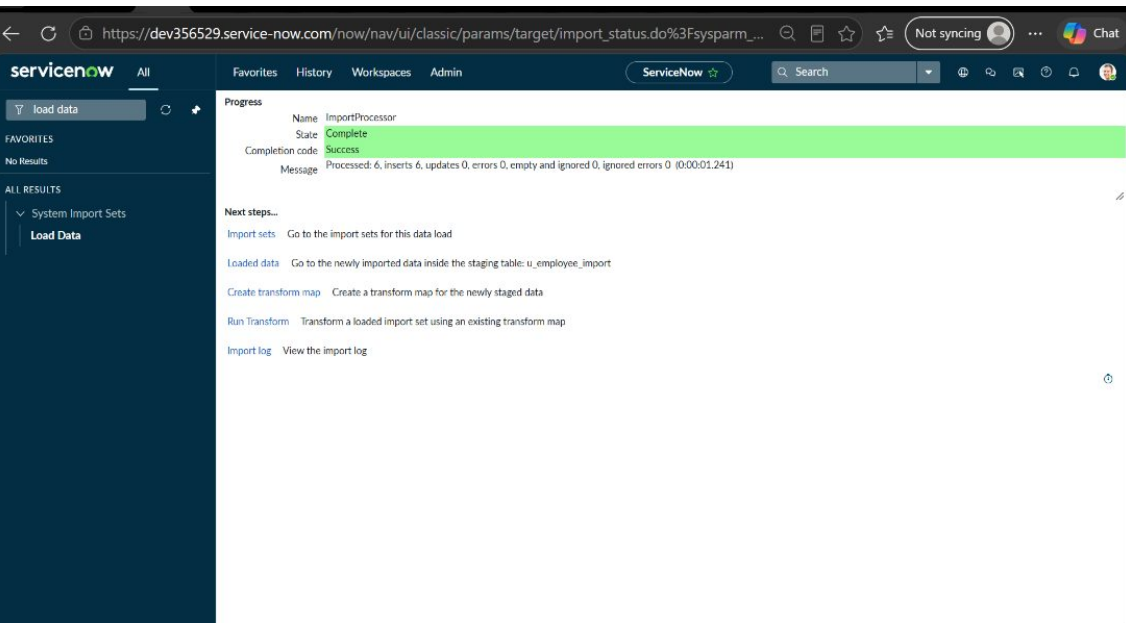
Step 7: Employee Test list after the first transform - 5 records (SB0001 to SB0005).



The screenshot shows the ServiceNow interface for the 'Employee Tests' list. The table contains 5 records with columns: Employee ID, Employee Name, Department, Email, and Location. The records are SB0002, SB0005, SB0004, SB0001, and SB0003. The interface includes a search bar, a filter dropdown, and a 'New' button. The bottom of the screen shows a Windows taskbar with the date 30-09-2026 and time 13:02.

| Employee ID | Employee Name | Department | Email           | Location  |
|-------------|---------------|------------|-----------------|-----------|
| SB0002      | user2         | servicenow | user2@gmail.com | Tamilnadu |
| SB0005      | user5         | Salesforce | user5@gmail.com | Tamilnadu |
| SB0004      | user4         | servicenow | user4@gmail.com | Tamilnadu |
| SB0001      | user1         | Servicenow | user1@gmail.com | Tamilnadu |
| SB0003      | user3         | servicenow | user3@gmail.com | Tamilnadu |

Step 8: Re-import with Load Data - State Complete, Success; 6 rows processed and inserted into the staging table.



The screenshot shows the ServiceNow 'Load Data' progress screen. The 'Progress' section indicates that the 'ImportProcessor' is in a 'Complete' state with a 'Success' completion code. The message states: 'Processed: 6, inserts 6, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.241)'. The 'Next steps...' section provides instructions on how to proceed with the data load.

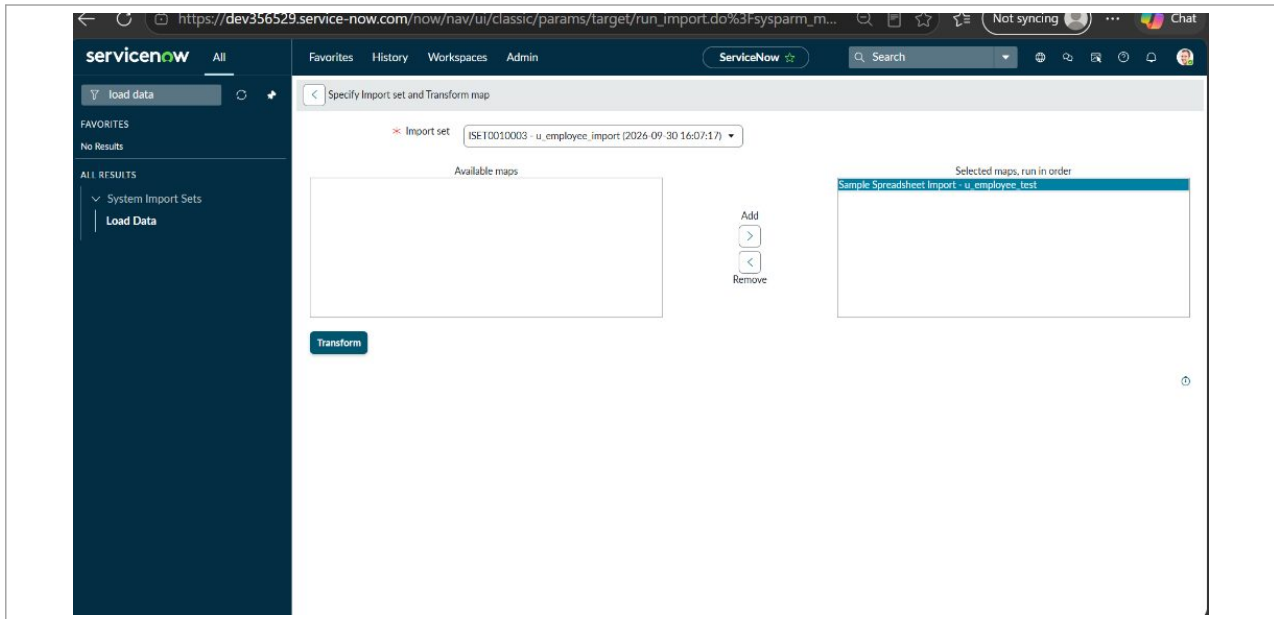
**Progress**

Name: ImportProcessor  
State: Complete  
Completion code: Success  
Message: Processed: 6, inserts 6, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.241)

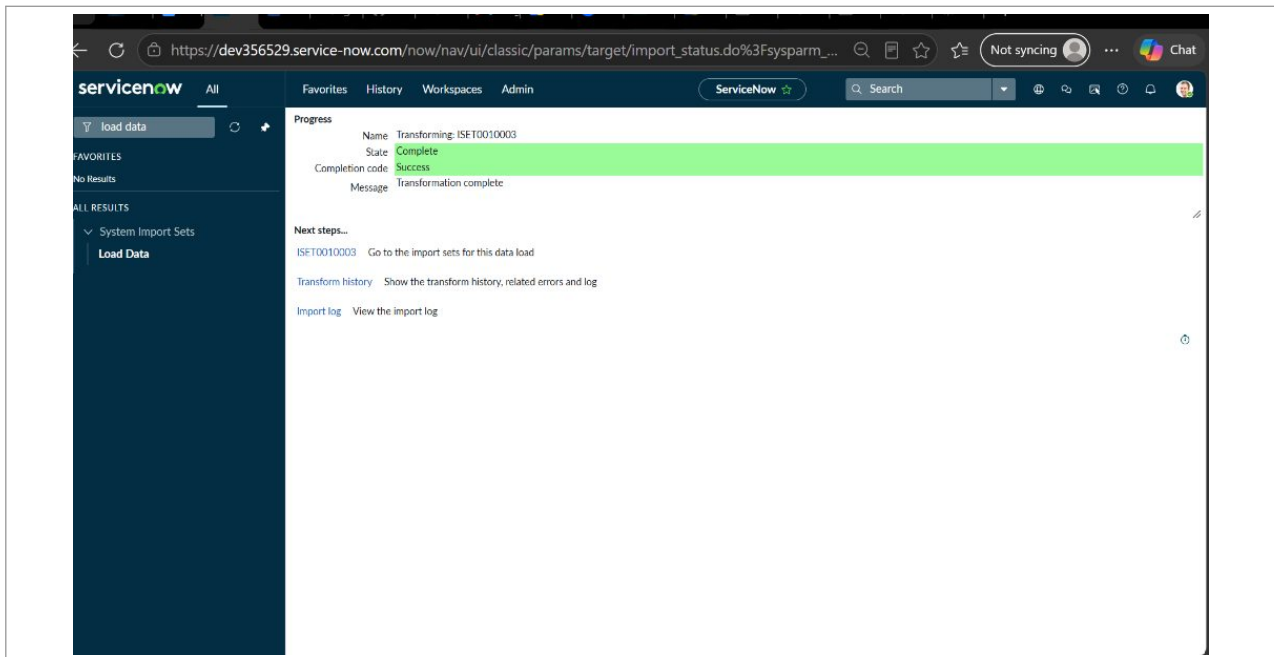
**Next steps...**

- [Import sets](#) Go to the import sets for this data load
- [Loaded data](#) Go to the newly imported data inside the staging table: u\_employee\_import
- [Create transform map](#) Create a transform map for the newly staged data
- [Run Transform](#) Transform a loaded import set using an existing transform map
- [Import log](#) View the import log

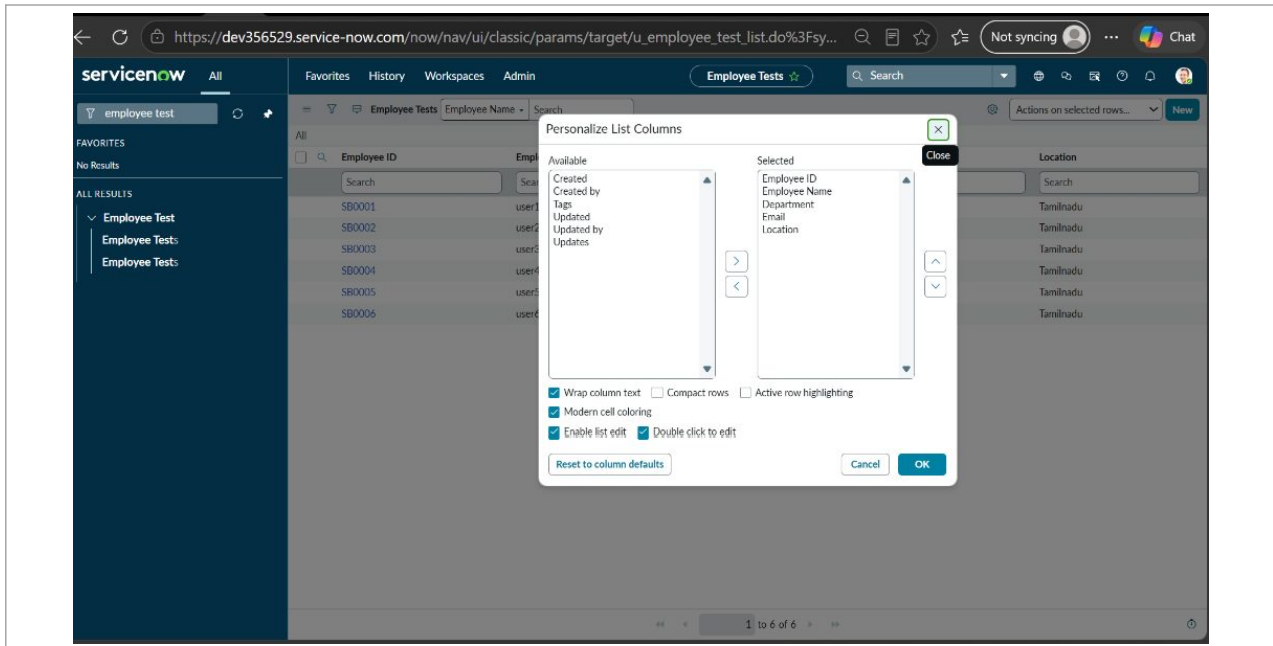
Step 9: Run Transform - import set ISET0010003 with the Sample Spreadsheet Import map selected.



Step 10: Transformation complete for ISET0010003 - State Complete, Success.



Step 11: Employee Test list with all 6 records, columns arranged using Personalize List Columns.



## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Automates bulk import of employee data and removes repetitive manual entry.
- Staging table protects the live table until data is mapped and transformed.
- Coalesce on Employee ID updates existing records and prevents duplicates.
- Transform History gives an audit of inserted, updated and ignored rows.
- Reports and a dashboard give real-time visibility, with no custom code required.

### Disadvantages

- Result quality depends on the quality of the spreadsheet; invalid rows must be corrected first.
- Coalesce relies on a unique, correctly entered Employee ID.
- Import is manual (load then transform) - it is not scheduled in this project.
- All fields are String type, so department and location are not linked to reference tables.
- Creating dashboards needs Admin Override on the pa\_dashboards ACL, which needs administrator rights.

## 9. CONCLUSION

This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency and data integrity throughout the employee onboarding and update process.

A key enhancement was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system identifies existing employee records during every import. This prevents duplicate records and ensures repeated imports update existing data instead of inserting redundant entries, keeping the employee database clean, reliable and consistent - critical where data may be imported multiple times from external systems.

The project was also extended with Reports and Dashboards to give real-time visibility into employee data and import activity. Custom reports track total employee records, newly added employees, updated employee records, department-wise distribution, and import success and error counts. These reports were consolidated into a dashboard so administrators and HR teams can monitor data trends, import performance and data quality from one central view, reducing manual analysis and improving decision-making.

## 10. FUTURE SCOPE

- Schedule imports with a scheduled data source so files load automatically.
- Replace String fields with reference fields to department and location tables (or sys\_user).
- Add transform scripts and mandatory-field enforcement for stronger validation.

- Send notifications for failed or partially ignored imports.
- Extend dashboards with import success / error trends and role-based sharing.
- Add an integration path through the Import Set API for system-to-system loads.

## 11. APPENDIX

**Source Code:** Not applicable - configuration-only solution (tables, transform map, reports, dashboard).

**Dataset:** Sample Spreadsheet.xlsx - columns: Employee ID, Employee Name, Email, Department, Location.

### References

ServiceNow Docs - Coalesce (Import Sets):

[https://www.servicenow.com/docs/r/integrate-applications/system-import-sets/c\\_ImportSetCoalesce.html](https://www.servicenow.com/docs/r/integrate-applications/system-import-sets/c_ImportSetCoalesce.html)

ServiceNow Docs - Import Sets API (staging table, data source, transform map):

<https://www.servicenow.com/docs/r/application-development/servicenow-sdk/fluent-import-sets-api.html>

ServiceNow Docs - Import Sets key concepts: [https://docs.servicenow.com/en-US/bundle/tokyo-platform-administration/page/administer/import-sets/concept/c\\_ImportSetsKeyConcepts.html](https://docs.servicenow.com/en-US/bundle/tokyo-platform-administration/page/administer/import-sets/concept/c_ImportSetsKeyConcepts.html)

ServiceNow Docs - Create a transform map:

[https://docs.servicenow.com/bundle/tokyo-platform-administration/page/script/server-scripting/task/t\\_CreateATransformMap.html](https://docs.servicenow.com/bundle/tokyo-platform-administration/page/script/server-scripting/task/t_CreateATransformMap.html)

S2 Labs - Transform Maps, Field Mapping and Coalescing:

<https://s2-labs.com/servicenow-admin/transform-maps-field-mapping-coalescing/>